

Week 3	
Objectives	The aims for this week are: - Start getting intro matrices. - Identify the different equations of lines and planes. You should also try to solve the first questionnaire of Challenge 2 this week.
Session 7	As an introduction, start watching the first part of the video “Matrices and systems of equations”, where we define the concept of a matrix, and you will see an example of applications of matrix for digital image representation. The video continues with brief examples on how to compute sums and products of matrices. You can also watch the part where the determinant, minor and rank of a matrix are defined. Once we get there, let's continue with the PDF resources. Start reading now section 3 of the module “Elements of linear algebra and geometry”. Read carefully how matrices are defined and the different kinds of matrices: the concepts are very easy to grasp, but it is important to keep them in mind because you will need them in the next sessions. There are plenty of definitions and properties, and all of them are simple to understand, but be careful with the small details! Learn also their basic operations. You can follow this module using the CalcMe Guide. Read section 3 of this module, until subsection 4.3. In the subsections 4.1 and 4.2 you will compute explicitly determinants of order 2, 3 and 4, which are easy to understand. Section 4.3 is about the properties of determinants: it is important that you try to summarize them in order to understand them better.
Session 8	Next, read subsection 4.4, where we will use determinants to compute the inverse of a matrix. Read carefully the methods used in 4.5 to compute the rank of a matrix using determinants. We will finish section 4 with two examples on computing the rank of a matrix. Read examples 41 and 42 and check yourselves all the computations done.
Session 9	Read now section 5, which begins describing the different equations for writing the equations of a line in the plane. You'll probably know them already, but review them anyway. Try to do example 48 and check that it's correct. In subsection 5.2 we will see the different equations of the line in the space. Try to re-do example 49 and check the results. It is important that you know the different ways of writing lines, and how to go from every one of them to the others. Please, don't continue until you fully understand this. Section 5.3 is about the equations of the plane in space. Try to re-do example 50, and solve exercises 13 and 14.
Keep in mind	It is very different to fully solve an exercise yourselves than to just read and understand the solution. Try to do them by yourself! You'll probably need to re-do the last session. Don't worry: it is a good investment of your time. Remember that you have to solve the first questionnaire of Challenge 2 this week.